

Assignment_02: LaTeX Preamble

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Introduction

This document demonstrates a meaningful and complete LaTeX preamble with multiple packages categorized by their functionality.

1 Page Layout & Structure

geometry is used to set page margins and layout.

setspace controls line spacing.

parskip adds space between paragraphs.

2 Mathematics

amsmath enables advanced equations:

$$E = mc^2$$

amssymb provides symbols like ∞ , $\leq$, $\geq$.

3 Figures

graphicx is used to include images.

caption customizes captions.

4 Tables

`array`, `booktabs`, `multirow`, and `longtable` improve table formatting.

5 Colors

`xcolor` allows colored text: This is red text

6 References

`hyperref` creates clickable links: [Example Link](#)

`natbib` manages citations.

7 Lists

- First item
- Second item

8 Units

`siunitx` example: 10 kg

9 Code

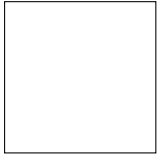
```
#include <stdio.h>
int main() {
    printf("Hello World");
}
```

10 Algorithm

Algorithm 1 Sample Algorithm

Start
Process
End

11 Graphics



12 Conclusion

This preamble supports equations, tables, figures, code, references, and formatting for professional documents.